



Nowcasting GDP using machine learning methods

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Abstract

This paper compares the ability of several econometric and machine learning methods to nowcast GDP in (pseudo) real-time. The analysis takes the example of Dutch GDP over the period 1992Q1–2018Q4 using a broad data set of monthly indicators. It discusses the forecast accuracy but also analyzes the use of information from the large data set of macroeconomic and financial predictors. We find that, on average, the random forest provides the most accurate forecast and nowcasts, whilst the dynamic factor model provides the most accurate backcasts.

Keywords Factor models · Forecasting competition · Machine learning methods · Nowcasting

JEL Classification C32 · C53 · E37

1 Introduction

GDP is published several weeks after the end of a quarter and initial releases are subject to substantial uncertainty. This fact led to the development of models to predict GDP of the current quarter, a practice referred to as "nowcasting". Such nowcasting models use dimension reduction techniques to nowcast GDP from a large number of macroeconomic and financial predictors.

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This paper explores the nowcasting precision of econometric and machine learning methods in comparison to the most commonly used nowcasting method, the dynamic factor model. We consider one and two quarters ahead forecasts, nowcasts, and backcasts of Dutch GDP using a broad data set of 83 macroeconomic and financial predictors. We consider the performance of several methods: factor models, regularization methods, random subspace methods, and the random forest. We separately evaluate their performance over different periods and different states of the economy.

For policy purposes, the interpretation of nowcasts is often of interest and we investigate how the different methods use the information from the large data set. This is relatively straightforward for the factor, regularization, and random subspace models. The random forest, in contrast, is highly nonlinear and we use Shapley values to calculate the importance of the different predictors.

Our findings suggest that, on average, the random forest provides the most precise predictions. An exception are the backcasts of the dynamic factor model until the financial crisis. Yet, this advantage of the dynamic factor model has disappeared and the random forest and LASSO model provide slightly more precise backcasts, since the financial crisis. We observe that the random forest and random subset selection use the different predictors considerably more evenly over categories as well as over horizons than the dynamic factor model and the LASSO.

An important benchmark for our analysis is the dynamic factor model as this model has been widely used at central banks to nowcast GDP (Giannone et al. 2008; Bańbura and Rünstler 2011; Jansen et al. 2016; Hindrayanto et al. 2016; Bok et al. 2018). Another popular class of models that we also consider are MIDAS based models (Marcellino and Schumacher 2010; Kuzin et al. 2011; Foroni and Marcellino 2014).

We compare these models to a range of models that are sometimes referred to as machine learning models. The first set of models are regularization methods, which estimate linear models, where the parameters are subject to penalization terms. The nature of the penalization distinguishes the different models. We consider the LASSO of Tibshirani (1996) and the elastic net of Zou and Hastie (2005). We also considered other penalization models, namely the adaptive LASSO and ridge regression. However, the results were inferior to the LASSO and elastic net and for brevity we omit them from the discussion. Babii et al. (2022) provide theoretical properties for the LASSO when estimating models with macroeconomic and financial time series data. They compare nowcasts of US GDP from the LASSO, ridge regression, and elastic net to the nowcasts of the New York Fed. They find that only their suggested specification of the LASSO outperforms the dynamic factor model of the Fed. This corresponds with our results for the Netherlands, where the LASSO and the elastic net are of comparable precision to the dynamic factor model over the entire forecast period.

Next, we consider the random subspace methods of Elliott et al. (2013) and Boot and Nibbering (2019). These methods exploit the fact that model averaging tends to reduce the mean square forecast error by making a large number of predictions, where the model for each prediction combines subsets of the predictors in the data

set in a random manner. The predictions are then averaged to yield the final forecast. Our results show that this is a competitive approach to nowcasting.

Finally, we employ the random forest of Breiman (2001), which combines the model averaging feature of the random subspace methods with the nonlinear modeling inherent in regression trees. A downside of the random forest is that its nonlinearity complicates the interpretation of the role of the different predictors in the forecasts and we work with Shapley values as discussed by Štrumbelj and Kononenko (2014) and Lundberg and Lee (2017) in our analysis of the importance of the predictors in the random forest.

We also consider averaging the forecasts from the different models using a range of weights that have been discussed by Elliott and Timmermann (2016). Model averaging has been shown to be able to reduce the mean square forecast error as it can reduce the variance of the resulting forecast. However, the weights used to average the forecasts introduce uncertainty, which can negatively impact the forecast precision. In our application, average forecasts are among the best forecasts over all horizons.

Our paper relates to the growing literature on forecasting with machine learning methods in macroeconomics. Richardson et al. (2018) evaluate nowcasts of New Zealand's GDP from a number of machine learning models using quarterly data. They find that machine learning methods, in particular support vector machines, and improve over their benchmark, the univariate autoregressive model. They do, however, not include the dynamic factor model or random forest, which are the leading methods in our analysis. Jönsson (2020) evaluates the nowcasting performance of the nearest neighbor algorithm for Swedish GDP and finds that it compares well to standard linear sentiment index models that are commonly used in Sweden. Yoon (2022) nowcasts Japanese GDP using boosted trees and random forests and finds that these outperform nowcasts of the Bank of Japan and the International Monetary Fund. Finally, Jardelet and Meunier (2022) nowcast world GDP using factor-augmented and LASSO MIDAS models. They find that factor-augmented models deliver the most precise nowcasts. An important difference between these papers and our work is that their analysis is restricted to average forecast performance measures. In contrast, the importance of the different predictors in the nowcasts is not considered. In a policy context, however, such interpretations are important and for this reason we put particular emphasis on the role of the predictors in our nowcasts.

Machine learning methods have also been used to predict other macroeconomic predictors. Medeiros et al. (2021) use a range of methods to predict U.S. inflation between 1 and 12 month into the future. Similar to our findings, their results suggest that the random forest is the most precise method to forecast inflation.

Gogas et al. (2021) predict euro area unemployment using machine learning methods and similarly find that the random forest delivers the most precise unemployment forecasts. Finally, Maehashi and Shintani (2020) forecast seven Japanese macroeconomic time series (but not GDP). Comparable to our results, they find that machine learning models tend to offer more precise forecasts than traditional time series models and that ensemble methods improve over individual models.

In the next section, we introduce the models that we use for nowcasting and discuss the interpretation of the role the different predictors play in the nowcasts.

Details of the data are in Sect. 3 and the results are discussed in Sect. 4. Finally, Sect. 5 concludes.

2 Nowcasting methodology

2.1 Nowcasting models

In this section, we give a brief overview of the different methods we use in this paper and the parameter choices we make. Since, these methods are well established in the literature, we relegate a more formal description to the online appendix,¹ We compare the nowcasts from these methods to those from two simple models. The first is the prevailing mean model, which takes the mean in the estimation sample as the forecast. This model serves as the benchmark in that we report the root mean square forecast error (RMSFE) of each method as a ratio to that of the prevailing mean. The second is the autoregressive model with lag length chosen by AIC with a maximum lag number of $p = 4$. All parameters are re-estimated for each nowcast using an expanding window.

2.1.1 Dynamic factor model

The dynamic factor model is widely used in the nowcasting literature. In this paper, we use the model developed by the Dutch central bank, which relies on the specification introduced by Giannone et al. (2008), Bańbura and Rünstler (2011), Jansen et al. (2016), and Bok et al. (2018). The nowcasting dynamic factor model is based on the approximate factor model of Stock and Watson (2002), which uses principal components to estimate the factor loadings in the dynamic factor model. The specification depends on the number of static factors (r), the number of dynamic factors (q) and the number of lags (p) in the VAR. In line with the literature (Kuzin et al. 2013; Jansen et al. 2016), we produce nowcasts for models using all combinations of one to six static factors, one to six lags for the static factors, and one to six dynamic factors (where $q \leq r$). In total we estimate 126 model specification. We then combine these nowcasts using an equal weighted average to obtain the final nowcast of the dynamic factor model.

2.1.2 Mixed-data sampling factor-augmented model

The MIDAS model of Ghysels et al. (2007) has been adapted for nowcasting by Marcellino and Schumacher (2010). In our factor-augmented MIDAS model, factors are extracted at a monthly frequency using PCA. In the unrestricted MIDAS model, which in our analysis was more precise than using exponential Almon lags, the monthly series of factors are included separately using skip sampling to model

¹ The DFM model is coded in Matlab and all other models in R, where all public code is available on <https://github.com/jasperdewinter/KantPickdeWinter>.

the quarterly series. In order to keep the model parsimonious, we extract between one and six factors from the regressor set, which includes the lagged regressors up to and including lag 6, which is a strategy that was shown to work well empirically by Pick and Carpay (2022). We then average the nowcasts obtained using the different number of factors. We then average the different forecasts. For brevity, we will refer to this factor-augmented MIDAS model as the MIDAS model.

2.1.3 Regularization techniques

We use the least absolute shrinkage and selection operator (LASSO) and the elastic net in this paper. These nowcasting models relate GDP growth to the full set of macroeconomic and financial predictors and their lags, and we skip sample these predictors to account for their monthly frequency. This leads to an extremely highly parameterized model and the necessary dimension reduction is then achieved through penalization.

Compared to the previous two models, the models using regularization assume a sparse DGP rather than the dense specification of the factor models. The difference between the two models is that the LASSO of Tibshirani (1996) performs both regularization and predictor selection by imposing an ℓ_1 penalty in the estimation of the coefficients, whereas the elastic net of Zou and Hastie (2005) imposes the ℓ_1 norm to select predictors and shrinks the remaining coefficients towards zero through the use of the ℓ_2 norm.

The amount of shrinkage in the LASSO is determined by a scalar parameter, λ , and by two scalar parameters, λ and α in the elastic net. We determine these parameters via cross-validation. Importantly, the cross-validation is done for each forecast separately and uses only data in the respective estimation sample, and this is the case for all empirical methods that use cross-validation.

2.1.4 Random subspace regression

Random subspace methods encompass methods that reduce the predictor space by averaging forecasts from either random combinations or random subsets of the data. The idea of randomly selecting smaller models and averaging (with equal weights) their forecast is based on complete subset regression of Elliott et al. (2013). They build on the idea of model averaging to combine forecasts obtained from all possible combinations of smaller linear models that can be produced from a large data set. However, the number of possible combinations can quickly become prohibitively large. A solution is to take a smaller number, say R , of randomly chosen subsets of predictors. Boot and Nibbering (2019) show that this approximates the complete subset regression for finite R , such as $R = 1000$.

The full predictor set contains all skip-sampled predictors and their lags. These are then sampled at random and a forecast, nowcast, or backcast is produced from a linear model containing the sampled predictors. The final prediction is the equally weighted average of the R predictions.

A tuning parameter of this method is the size of each predictor subset, k . Theoretical results by Boot and Nibbering (2019) suggest that k should be chosen relatively

large at about 30. The experience of Pick and Carpay (2022) suggests that smaller k can deliver more precise forecasts. We initially experimented with different choices of k up to 30 and our experience confirms that smaller choices of k deliver better nowcasts. As a result, we average nowcasts over those obtained using $k = 2, 3, 4, 5$.

An alternative to selecting predictors is to use weighted averages of the predictors. Boot and Nibbering (2019) discuss this option and call it ‘random projection’. The skip-sampled predictors and their lags are combined to a small number k of weighted averages using a random weighting matrix. For Gaussian random projections, the weights are independently drawn from a standard normal distribution. The k weighted averages are then used in a linear regression model to predict GDP growth. R realizations of the weights are drawn and the resulting forecasts are averaged with equal weights. Again, k needs to be determined and, as above, we average nowcasts using $k = 2, 3, 4, 5$.

Other weighting schemes are possible, for example, the weights used by Guhaniyogi and Dunson (2015). The analysis of Pick and Carpay (2022), however, suggests that random subset selection and random projection deliver superior forecast performances and we therefore limit our attention to these two methods.

2.1.5 Random forest

The random forest averages the forecasts of multiple regression trees. To grow a regression tree, the space of predictors, which, similar to above, are skip-sampled, is partitioned with the aim of minimizing the in-sample squared error. At each partition, the algorithm chooses a split based on one of the predictor that realizes the largest decrease in squared error.

Trees are designed to have a high degree of independence of each other by randomly drawing a subset of predictors and a subset of observations to grow any given tree. Averaging the forecasts from the trees in the random forest therefore minimizes the variance of the average forecast.

Regression trees, if unchecked, have the tendency to overfit the data. In order to reduce overfitting, each estimation sample is divided in a training and a validation set. The share of the data in the training set, ω , in all estimation samples is varied with $\omega = 0.6, 0.7, 0.8, 0.9$ and we average forecasts over the results from the ω . Within each training set size, ω , we cross-validate the number of skip-sampled predictors to grow the tree, where the possible values of the number of predictors is $k = 1, 2, \dots, 249$. We use the prediction of the 400 trees in each random forest.

2.1.6 Remarks

The dynamic factor model and unrestricted MIDAS model assume a dense DGP in the sense that all variables contribute to the estimation of the factors. The dimension reduction in these methods does not target GDP exclusively but yields the factors summarize the information of the entire data set. The regularized models, in contrast, assume a sparse data structure, which is reflected in the exclusion of many variables through the ℓ_1 norm based penalization. They also target GDP directly.

Random subset and random forest, finally, suggest a dense data structure. Like regularization methods, they target the variable of interest directly.

Each method requires hyperparameters as input and Table 1 summarizes these choices. We attempt to limit the effect of a single hyperparameter and use forecast averaging over many of the hyperparameters. The exception is the regularization methods where, in line with the literature, we choose the hyperparameters that minimize the loss function in the cross-validation sample.

2.2 Nowcast combinations

Given that the models above have distinct ways of incorporating the information of the monthly indicators, combining the forecasts could be beneficial. Combining several forecasts from different sources has a long track record in the forecasting literature (Timmermann 2006). Forecast combinations require combination weights. As optimal weights can be determined only under very specific assumptions, we use a range of practical approaches to determine the weights.

The first, simple solution is to give the forecasts equal weight, which turns out to be a difficult to beat benchmark (Timmermann 2006). An advantage of equal weighted forecast combinations is that fixed weights avoids estimation uncertainty that would translate into forecast uncertainty. The downside, however, is that if the unknown optimal weights are far from equal weights, the forecast combination may suffer. We therefore also consider ways to estimate the weights.

Table 1 Hyperparameter choices

Method	Hyperparam	Choice
DFM	No. static factors, r	Ave. nowcasts over $r = 1, 2, \dots, 6$ factors
	No. dynamic factors, q	Ave. nowcasts over $q = 1, 2, \dots, 6$ dyn. factors
	Lag length, p	Ave. nowcasts over $p = 1, 2, \dots, 6$ lags
MIDAS	No. static factors, r	Ave. nowcasts over first $r = 1, 2, \dots, 6$ factors From regressor set with 6 lags
LASSO	Weight ℓ_1 penalty, λ	Cross-validation
EN	Weight ℓ_1 penalty, λ	Cross-validation
	Weight ℓ_1 vs ℓ_2 penalty, α	Cross-validation
RS	Size models, k	Ave. nowcasts over $k = 2, 3, 4, 5$
	No. of models, R	1000
RP	Size models, k	Ave. nowcasts over $k = 2, 3, 4, 5$
	No. of models, R	1000
RF	Share training set ω	Ave. nowcasts over $\omega = 0.6, 0.7, 0.8, 0.9$
	No. of predictors per tree, κ	Cross-validate per ω for $\kappa = 1, 2, \dots, 249$
	No. of trees	400

The methods are: DFM the dynamic factor model, MIDAS the mixed-data sampling factor-augmented model, LASSO the least absolute shrinkage and selection operator, EN the elastic net, RS the random subset selection, RP the random projection, RF the random forest. ‘ave. nowcasts’ denotes that we average the nowcasts over the given choices of hyperparameters

The first method to estimate the forecast weights is to use weights that are inversely proportional to the MSFE. We measure the cumulative square forecast error, $v_{j,t|t-h}^2$, and calculate the weights as

$$w_{j,t} = \frac{(v_{j,t|t-h})^{-2}}{\sum_{j=1}^m (v_{j,t|t-h})^{-2}}$$

where m is the number of forecasts to combine. We calculate the cumulative error, $v_{j,t|t-h}$, using either an expanding window or a rolling window of ten quarters. Prior to the first forecast, no weights can be determined and we take an equally weighted average as the first forecast combination.

The weights above do not address potential biases of the forecasts. If biases are suspected, Granger and Ramanathan (1984) suggest estimating weights in a linear regression

$$y_{t+h|t} = \beta_0 + \sum_{j=1}^m \beta_j \hat{y}_{j,t+h|t} + \epsilon_{t+h}$$

where the estimated coefficients are then used as forecast weights in addition to the intercept that estimates the bias. Again, we estimate the coefficient using an expanding and a rolling window of size 40. For the first 40 forecasts we use equal weights.

2.3 Nowcast evaluations

We report the forecast performance as measured by the root mean square forecast error (RMSFE) of each method relative that of the prevailing mean forecast. Define the forecast error as $\hat{e}_{t|t-h,a} = y_t - \hat{y}_{t|t-h,a}$, where subscript a denotes the respective method. Then the ratio of RMSFE is

$$\text{relative RMSFE} = \frac{\sqrt{\frac{1}{T_f} \sum_{t=1}^{T_f} \hat{e}_{t|t-h,a}^2}}{\sqrt{\frac{1}{T_f} \sum_{t=1}^{T_f} \hat{e}_{t|t-h,pm}^2}}$$

where subscript pm the prevailing mean forecast, and T_f denotes the number of forecasts. For the prevailing mean, we report the levels of the RMSFE.

In order to distinguish the effect of bias and standard deviation on the forecast performance, we also report the ratio of the absolute forecast bias and the ratio of the forecast standard deviations. The ratio of the absolute bias is

$$\text{relative abs.bias} = \frac{\frac{1}{T_f} \sum_{t=1}^{T_f} |\hat{e}_{t|t-h,a}|}{\frac{1}{T_f} \sum_{t=1}^{T_f} |\hat{e}_{t|t-h,pm}|}$$

and the ratio of the standard deviation is

$$\text{relative std.deviation} = \frac{\sqrt{\frac{1}{T_f} \sum_{t=1}^{T_f} \hat{e}_{t|t-h,a}^2 - \bar{\hat{e}}_{h,a}^2}}{\sqrt{\frac{1}{T_f} \sum_{t=1}^{T_f} \hat{e}_{t|t-h,pm}^2 - \bar{\hat{e}}_{h,pm}^2}}$$

where $\bar{\hat{e}}_{h,a} = \frac{1}{T_f} \sum_{t=1}^{T_f} \hat{e}_{t|t-h,a}$ for a given nowcasting method a , including the prevailing mean model.

We also use the test of Diebold and Mariano (1995) to evaluate significance of the forecasts of the different models against those from the dynamic factor model. The Diebold-Mariano test results should be interpreted with caution, since we are using an expanding window, which could imply that the assumptions of the Diebold-Mariano test are violated. Significance can therefore be interpreted as a sign of improved forecast performance relative to insignificant forecasts but not necessarily at the stated significance level.

2.4 Interpreting nowcasts

In policy environments, such as central banks, the interpretation of nowcasts is important. We will therefore illustrate the contributions of the underlying time series to the nowcasts over time. Note, however, that the contributions merely provide the importance of predictors in a given model, which need not imply a structural economic interpretation. For the dynamic factor model, the weights that, in a given model, are assigned to predictors are determined using the methods of Koopman and Harvey (2003), see also Bańbura and Rünstler (2011).²

The MIDAS, LASSO, elastic net, random subset regression, and random projection are all linear in the predictors. This means that it is straightforward to extract the contributions of the time series to the forecasts. For the shrinkage methods, the contributions of the time series to the forecasts follow from the linear relationship the selected predictors have with GDP. The random subspace methods also fall into the class of linear models conditional on the selection predictors and the projection matrices.

The random forest, in contrast, is a highly nonlinear model, which makes interpreting the role of different predictors considerably more complex. We use the concept of Shapley values (Shapley 1953) to interpret the role of predictors, which has been developed further by Štrumbelj and Kononenko (2014) and Lundberg and Lee (2017).

The Shapley value measures the average difference in the loss of trees that include the predictor in question to the loss of trees that do not include that predictor. Denote the loss of a tree that includes a given predictor i by $L(S \cup \{i\})$ and that of another tree with the same predictors except predictor i by $L(S)$, where S is the

² We assign the contribution of the constant to the five predictor groups based on the derived average weight per predictor group. We calculate weights for each forecasting horizon and re-scale all weights to lie in the interval $[0.01, +\infty)$. See Bańbura and Rünstler (2011) for analytical derivation of the weights.

set of predictors in the tree except predictor i , and $S \subseteq \mathcal{F}$ with $\mathcal{F}$ denoting the complete set of predictors. As the effect of predictor i likely depends on the other predictors in the tree, the loss differential is computed for all possible predictors in a tree. The contribution of predictor i , ϕ_i , is the weighted average of the loss differences

$$\phi_i = \frac{1}{N_M} \sum_{S \subseteq \mathcal{F}} [L(S \cup \{i\}) - L(S)]$$

where N_M is the number of possible combinations of predictors in trees excluding predictor i . With many predictors, this procedure is computationally burdensome and we therefore use the approximation proposed by Štrumbelj and Kononenko (2014), which uses randomly sampled subsets of predictors to compute the Shapley value for each predictor.

3 Data

The data set consists of 83 monthly time series and quarterly GDP that were downloaded on the 26th of March 2019. The statistical monthly information set reflects the public knowledge at the end of the month. The time series were obtained from Statistics Netherlands, the central banks of Belgium and the Netherlands, Datastream, the European central bank, Eurostat, and the Hamburg Institute of International Economics and the Dutch RAI association.

The series can be grouped into five categories. The first category is hard, quantitative information on production and sales, such as industrial production, car sales, retail sales, exports, imports, and unemployment. The second category is soft, qualitative information on expectations derived from surveys among consumers, retailers, and firms. The third category contains financial predictors, both quantities (monetary aggregates) and prices (interest rates and stock prices), which determine financing conditions for firms and consumers. Moreover, financial market prices partly reflect financial market expectations on output developments in the near future. The fourth category contains information on prices, i.e. consumer prices, producer prices, the housing price, and commodity prices. The fifth category contains some miscellaneous series, i.e. bankruptcies and issued vehicle registrations.

Most monthly data are seasonally (and calendar effects) adjusted at the source, except for prices and financial predictors. If data are not seasonally adjusted we apply the US Census' X12-method. All monthly series are made stationary by differencing, log-differencing or double log-differencing (in the case of prices). Moreover, all predictors are standardized by subtracting the mean and dividing by the standard deviation. This normalization is necessary to avoid overweighting of high-variance series in the extraction of common factors. Seasonal adjustment and standardization is done in pseudo-real time so as to mimic the process of a nowcaster over our forecast sample. Details can be seen in Table A.2 in the online appendix.

All monthly indicator series start in January 1985, while the quarterly GDP series start in the first quarter of 1985. The estimation period starts in 1986M1. The forecast evaluation period runs from 1992Q1 to 2018Q4. We produce

pseudo-out-of sample forecasts for the observations in the forecast period using an expanding estimation sample, that is, model selection and the estimation of the parameters in the models relies on the data of the estimation sample and, after each forecast, the estimation sample is expanded to include the additional observations.

Our forecast period spans different economic episodes. We, therefore, separately analyze the forecasts for the period until the financial crisis, that is, the period of the great moderation (1992Q1–2008Q1). Next, we consider the Financial Crisis (2008Q2–2011Q1) and, finally, the period post financial crisis (2011Q2–2018Q4). We abbreviate these periods as GM, FC and PFC, respectively. Additionally, we analyze the performance of the model during recessionary periods, expansionary periods and periods of moderate growth, based on the [OECD](#) Recession indicator for the Netherlands. Based on our expert judgment we remove the 2001 recession and the 2018 recession from the mechanical business cycle dating tool of the OECD because these periods could better be characterized as periods of moderate growth than recessions. Moreover, we characterize the period between the 2008–2009 recession and the 2011–2013 recession as a period of moderate growth as well. During the latter period, growth remained stagnant. Over our entire forecasting period, the Dutch economy was in recession during 22 quarters, expanded during 57 quarters, and during 29 quarters the economy grew moderately. Figure 1 shows the periods graphically.

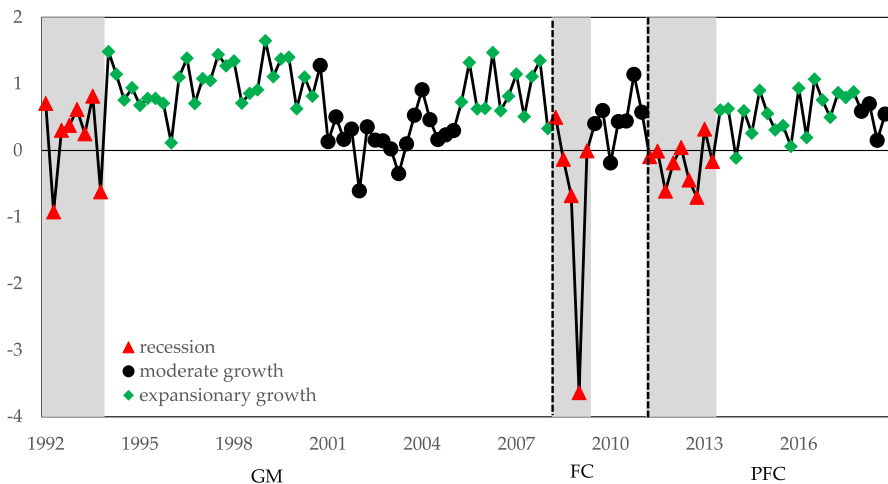


Fig. 1 Real quarter-on-quarter GDP growth in the Netherlands, 1992Q1–2018Q4. Note: Three time periods are depicted: GM: great moderation (1992Q1–2008Q1), FC: financial crisis (2008Q2–2011Q1) and PFC: post-financial crisis (2011Q2–2018Q4). Shaded areas show recession periods according to OECD recession indicator for the Netherlands

3.1 The ragged edge

The different indicators are published with varying delays. Generally, the hard predictors are accompanied with a significant publication delay, whereas soft predictors (surveys) are published on a timelier basis. The varying availability of the indicators is commonly referred to as the "ragged-edge" (Wallis 1986).

Our modelling strategy accounts for data availability at the moment when the prediction would be made in real time. Our approach is akin to the direct forecasting approach: rather than forecast predictors that are still unreleased and include the predicted values in our nowcasts, we predict GDP using the latest available data. *M1*, for example, has a publication delay of two month (see Table A.2 in the online appendix). Predictions for all horizons in May, say, will therefore use the *M1* number from March. The estimation of each model equally account for the publication lags such that treatment of publication lags is consistent throughout the modelling. This has been called "vertical realignment" by Marcellino and Schumacher (2010). We have experimented with other methods to deal with the ragged-edge, such as using univariate methods to predict the missing observations. However, these methods, while computationally much more costly, did not lead to improvements in nowcast precision and, for brevity, we therefore omit these results.

We employ a pseudo real-time design, which takes data publication delays into account, but ignores the possibility of data revisions for GDP and some indicators, such as retail trade. The latter might imply that we overestimate the forecasting accuracy of statistical models. However, it is also quite likely that the effects of data revisions on the final forecast will largely cancel out because statistical methods typically attempt to eliminate noise from the process by either extracting factors from a large data set or pooling large numbers of indicator-based forecasts. For example, using real-time data vintages for Germany, Schumacher and Breitung (2008) did not find any clear impact of data revisions on the forecast errors of factor models. Moreover, the effect on the relative performances of models, which is the main focus of this paper, is likely to be quite small (see Bernanke and Boivin 2003).

4 Results

4.1 Model performance

Table 2 reports the RMSFE over the entire forecast sample over eleven forecast horizons: three monthly forecasts for each of the two quarters before the target quarter, three monthly nowcasts during the target quarter, and two monthly backcasts in the quarter after the target quarter. It can be seen from the table that for the short-term forecasts and nowcasts, only the random forest beats the prevailing mean at all horizons. It is also the most precise forecast for all the short term forecasts and the first two nowcasts. The Diebold-Mariano test shows statistical significance for these horizons. For the third nowcast, the LASSO is slightly more precise. The dynamic factor model is the most precise method when backcasting.

Table 2 Average forecast precision over the period 1992Q1–2018Q4

Horizons	Benchmarks			Alternatives					
	PM	AR	DFM	MIDAS	LASSO	EN	RS	RP	RF
2Q.1	0.68	1.04	1.02	1.06	1.00	1.01	1.00	1.02	0.97*
2Q.2	0.68	1.04	1.02	1.04	1.00	0.99	1.02	1.02	0.95**
2Q.3	0.67	1.03	1.01	1.07	0.99	0.99	0.99	1.01	0.92**
1Q.1	0.67	1.03	1.00	1.04	0.98	0.99	1.00	1.01	0.94**
1Q.2	0.67	1.03	0.99	1.03	0.98	0.99	0.99	1.00	0.95*
1Q.3	0.67	1.02	0.99	1.02	1.01	1.00	0.97	0.99	0.92**
N.1	0.67	1.01	0.98	1.00	0.94*	0.95*	0.95**	0.99	0.91**
N.2	0.67	1.01	0.95	1.00	0.93	0.92	0.93	0.97	0.90**
N.3	0.67	0.96	0.88	1.00	0.87	0.87	0.90	0.94	0.88
B.1	0.67	1.00	0.83	0.97	0.84	0.86	0.91	0.96	0.89
B.2	0.67	1.00	0.79	0.97	0.82	0.87	0.90	0.96	0.88

Note: The table reports the results over the entire forecast sample. The first column reports the periods of origin of the forecast: $XQ.Y$ denotes a X quarters ahead forecast made in the Y -th month in the quarter, $N.Y$ denotes nowcasts made in the Y -th month of the quarter, and $B.Y$ denote backcasts made in the Y -th month of the quarter. For the prevailing mean (PM) the entries (in italics) refer to the level of the RMSFE; for all other models the entries refer to the RMSFE relative to the RMSFE of the PM model. Bold values indicate the model with the lowest RMSFE. AR denotes the autoregressive model, DFM the dynamic factor model, MIDAS the mixed-data sampling factor-augmented model, LASSO the least absolute shrinkage and selection operator, EN the elastic net, RS the random subset selection, RP the random projection, RF the random forest. *, **, *** indicate statistical significance at 10, 5 and 1 percent levels in a one-sided Diebold-Mariano test relative to the DFM (the significance levels should be interpreted with caution due to the use of expanding windows)

The AR and random projection are generally no more precise than the dynamic factor model. The MIDAS model is less precise than the DFM at all horizons. The LASSO, elastic net, and random subset regression are slightly more precise than the dynamic factor at most forecasting and nowcasting horizons but less precise when backcasting. This suggests that over the entire forecast sample, the non-linearity of the random forest does well when fore- and nowcasting. Other methods that rely on linear combinations of the data cannot substantially improve on the dynamic factor model. For backcasting, the dynamic factor model clearly is the most precise approach.

Figure 2 shows the levels of RMSFEs, which generally decrease as more information becomes available. The random forest has consistently lower RMSFE than the other methods for the forecasting months. For the first two nowcasting months it is also the most precise. During the last nowcasting month, the regularization methods and the dynamic factor model catch up with the random forest. In the backcasting months the dynamic factor model improves substantially over the other methods. Of the other models, the AR model, MIDAS and random projection methods remain at the upper level of the RMSFE range.

The forecast sample contains periods of different nature, such as the Great Moderation and the Financial Crisis. In order to evaluate the robustness of our

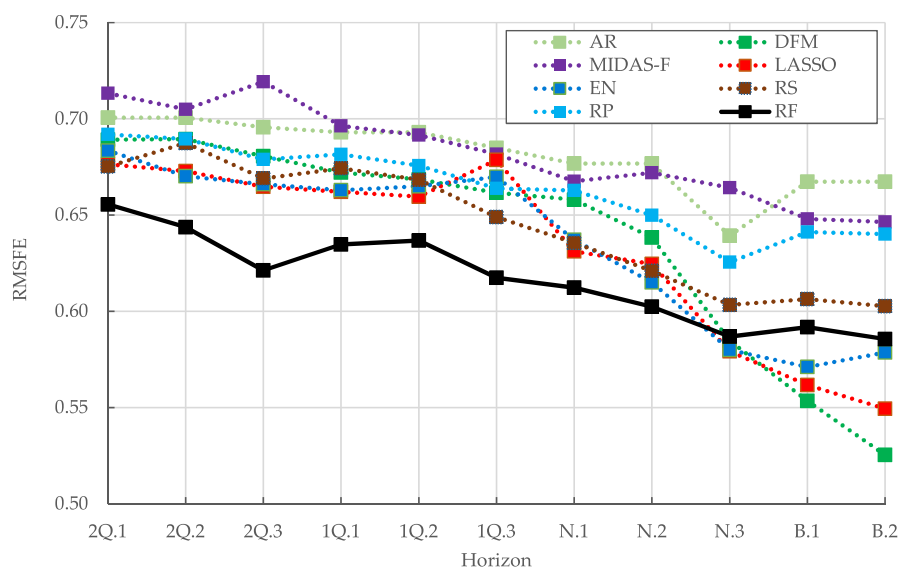


Fig. 2 RMSFE over varying forecast horizons. Note: The table reports the level of the RMSFE of the different methods. For further details see the footnote of Table 2

results, Table 3 reports the (ratio of) RMSFE over three subperiods: the Great Moderation, Financial Crisis, and the Post-Financial Crisis period.

During the Great Moderation, the two quarter ahead forecast from all models, with the exception of the AR model, are roughly equivalent in precision to that of the benchmark model for the first two months. In the last month of the two-quarter ahead forecast, the random forest is substantially better than the benchmark and significantly better than the DFM. As we get closer to the target quarter, the random forest continues to provide the most precise predictions one quarter ahead and also the most precise nowcasts by a substantial margin. The most precise backcasts, however, come from the dynamic factor model. The AR and MIDAS models fail to beat the prevailing mean benchmark at any horizon.

During the financial crisis, the prevailing mean forecast is substantially less precise in absolute terms. Still, most models fail to deliver more precise one and two quarter ahead forecasts than this benchmark. The exceptions are the random forest and, to a lesser extent, the MIDAS models. The random forest is more precise than the prevailing mean over all forecasting horizons and offers improvements of up to 18% against the prevailing mean. For the now- and backcast all methods improve substantially over the prevailing mean with the LASSO being the most precise, with the dynamic factor model and elastic net trailing by a small amount. The gain in relative forecasting accuracy can amount to 35% against the prevailing mean.

Since the financial crisis, the random forest has the most precise forecasts, nowcasts, and first backcast with improvements of up to 13%. The DFM also improves over the benchmark when nowcasting and backcasting albeit to a smaller degree than the random forecast. The MIDAS and regularization methods fail to consistently

Table 3 Forecast precision evaluated over different subperiods

Horizons	Benchmarks			Alternatives					
	PM	AR	DFM	MIDAS	LASSO	EN	RS	RP	RF
<i>Great Moderation (N= 65)</i>									
2Q.1	0.53	1.06	1.00	1.09	1.00	1.03	1.00	1.05	1.03
2Q.2	0.53	1.06	1.00	1.06	1.01	0.99	1.05	1.02	1.00
2Q.3	0.53	1.05	0.99	1.06	0.95	0.95	1.00	0.98	0.92*
1Q.1	0.53	1.05	0.97	1.05	1.03	1.02	1.02	1.02	0.96
1Q.2	0.53	1.05	0.98	1.06	1.00	1.01	1.00	1.00	0.95
1Q.3	0.53	1.04	0.98	1.05	1.04	0.99	0.98	0.98	0.92*
N.1	0.53	1.04	0.98	1.02	1.00	1.01	0.97	1.03	0.94**
N.2	0.53	1.04	0.98	1.04	1.01	0.99	0.95	1.02	0.92**
N.3	0.53	1.02	0.95	1.04	0.99	0.99	0.97	1.00	0.91*
B.1	0.53	1.03	0.92	1.03	0.99	0.98	0.99	1.03	0.97
B.2	0.53	1.03	0.88	1.03	0.98	0.98	0.97	1.02	0.94
<i>Financial crisis (N= 12)</i>									
2Q.1	1.36	1.02	1.02	0.97**	1.00	0.99	0.99*	0.99*	0.92**
2Q.2	1.36	1.02	1.02	0.97*	0.97	0.97	0.99	1.00	0.90**
2Q.3	1.36	1.03	1.02	0.98	0.98	1.00	0.99	1.01	0.91**
1Q.1	1.36	1.02	1.02	0.99	0.95	0.97	0.99	1.00	0.94*
1Q.2	1.36	1.02	1.01	0.99	0.95	0.95	0.99	1.00	0.95
1Q.3	1.35	1.02	0.99	0.97	0.98	0.97	0.96***	0.99	0.92*
N.1	1.35	1.01	0.99	0.95	0.90**	0.92**	0.95***	0.96	0.90*
N.2	1.35	1.01	0.93	0.95	0.78*	0.79**	0.91	0.93	0.88
N.3	1.35	0.93	0.78	0.94	0.69**	0.71*	0.85	0.87	0.85
B.1	1.35	0.99	0.71	0.88	0.65	0.72	0.83	0.90	0.82
B.2	1.35	0.99	0.66	0.90	0.65	0.69	0.83	0.91	0.83
<i>Post financial crisis (N= 31)</i>									
2Q.1	0.52	1.03	1.04	1.19	1.02	1.02	1.03	1.05	0.95*
2Q.2	0.52	1.03	1.05	1.17	1.03	1.04	1.02	1.08	0.97
2Q.3	0.52	1.01	1.03	1.28	1.08	1.05	1.02	1.08	0.97
1Q.1	0.52	1.02	1.01	1.13	0.97	0.96	0.98	1.02	0.92**
1Q.2	0.52	1.02	0.97	1.06	1.00	1.04	0.99	1.02	0.92
1Q.3	0.52	0.98	0.99	1.07	1.04	1.11	0.97	1.02	0.92
N.1	0.52	0.96	0.98	1.06	0.93	0.92	0.92	0.99	0.92
N.2	0.52	0.96	0.96	1.06	1.12	1.05	0.92	0.98	0.88
N.3	0.51	0.89	0.93	1.03	1.00	0.98	0.90	0.96	0.87**
B.1	0.51	0.96	0.90	1.06	0.92	0.90	0.92	0.94	0.90
B.2	0.51	0.96	0.88	1.01	0.86	1.02	0.93	0.95	0.87

For further details see the footnote of Table 2

beat the benchmark over this period. The random subspace methods, in particular the random subset regression, provide substantial improvements when nowcasting and backcasting, even if the forecasts are not consistently better than the benchmark.

Table 4 reports the RMSFE in different economic states. During recessions, the prevailing mean has a relatively large RMSFE and all methods, with the exception of the AR and the MIDAS models, improve over it over all horizons. This holds especially for the late nowcasts and backcast, when the forecasting gains amount to 37% for the best model. The random forest provides the relatively most precise two-quarter ahead forecasts. The LASSO model is the most precise for the one quarter ahead forecasts and nowcast for most months. The dynamic factor model provides the most precise backcast.

Over periods of moderate growth, the random forest provides the most precise forecasts at all horizons. It is the only methods that consistently beats the prevailing mean benchmark. During expansions, the prevailing mean benchmark is considerably more precise than during the recessions and moderate growth. Most other methods provide similarly precise forecasts but cannot consistently beat the prevailing mean. Compared to the dynamic factor model, the random forecast provides more precise nowcasts, whereas the MIDAS model and the regularized methods provide considerably less precise nowcasts.

A question is whether the forecast precision of the different methods is due to an increase or reduction in biases or standard deviations over the benchmark. Table 5 displays (ratios of) the absolute forecast biases. The prevailing mean results show that the bias is the relatively smaller component for this benchmark forecast. The biases of the dynamic factor model, MIDAS, random subset, and random projection methods are comparable to or larger than that of the prevailing mean forecast. Only the LASSO, elastic net and the random forest can reduce the absolute bias with the random forest offering the largest reduction in bias on average.

Table 6 shows the (ratios of) the forecast standard deviations. In contrast to the results for the bias, all methods, except the AR model, tend to reduce the forecast standard deviation compared to the prevailing mean forecast. For the first eight horizons, the random forest offers the largest reduction by a substantial margin. For the nowcast, all methods except the AR model, offer substantial reductions. For the backcast, finally, the dynamic factor model reduced the forecast standard deviation by the largest amount. Overall, for most methods the improvements in MSFE over the benchmark can be explained by an improvement in this component. The random forest is the only method that offers substantial reduction in biases and standard deviation over all horizons.

4.2 Forecast combinations

While the random forest tends to provide the most precise forecasts over many economics states, this is not uniformly so. It may therefore be worth to consider forecast combinations, where we combine the seven models using dimension reduction. Table 7 reports the results of the forecast combination schemes. The equally

Table 4 Forecast precision over various economic states

Horizons	Benchmarks			Alternatives					
	PM	AR	DFM	MIDAS	LASSO	EN	RS	RP	RF
<i>Recessions (N= 22)</i>									
2Q.1	1.21	1.03	1.00	1.03	0.99	0.99	0.97	1.00	0.92 *
2Q.2	1.21	1.03	0.99	1.01	0.96	0.97	0.99	1.01	0.92 *
2Q.3	1.21	1.03	0.97	1.04	0.97	0.97	0.97	1.00	0.91 *
1Q.1	1.21	1.03	0.95	1.00	0.90	0.92	0.97	0.99	0.91**
1Q.2	1.21	1.03	0.94	0.98	0.88	0.90	0.94	0.98	0.92
1Q.3	1.20	1.02	0.95	0.97	0.92	0.91	0.93	0.97	0.89 **
N.1	1.20	1.00	0.93	0.96	0.88 *	0.89	0.91	0.95	0.89*
N.2	1.20	1.00	0.87	0.94	0.82	0.83	0.88	0.93	0.88
N.3	1.20	0.92	0.75	0.93	0.76	0.76	0.83	0.88	0.85
B.1	1.20	0.99	0.68	0.90	0.73	0.77	0.84	0.91	0.84
B.2	1.20	0.99	0.63	0.87	0.66	0.69	0.84	0.91	0.83
<i>Moderate growth (N= 29)</i>									
2Q.1	0.53	1.06	1.07	1.10	1.06	1.07	1.05	1.07	0.96 *
2Q.2	0.53	1.06	1.09	1.09	1.11	1.03	1.06	1.06	0.89 **
2Q.3	0.52	1.06	1.12	1.12	0.99	1.03	1.05	1.05	0.84 **
1Q.1	0.52	1.04	1.12	1.11	1.21	1.14	1.09	1.08	0.97
1Q.2	0.52	1.04	1.12	1.14	1.17	1.14	1.11	1.07	0.94
1Q.3	0.52	1.05	1.07	1.11	1.14	1.10	1.07	1.05	0.89 *
N.1	0.52	1.02	1.11	1.05	1.06	1.08	1.04*	1.08	0.94
N.2	0.52	1.02	1.14	1.12	1.00*	0.97*	1.04**	1.08	0.89 **
N.3	0.52	0.97	1.09	1.11	0.97	0.91**	1.06	1.06	0.84 **
B.1	0.52	1.02	1.07	1.11	0.96	0.95	1.02	1.07	0.90 *
B.2	0.52	1.02	1.05	1.14	1.02	1.03	1.03	1.08	0.89 *
<i>Expansionary growth (N= 57)</i>									
2Q.1	0.39**	1.03	1.06	1.09	0.99 ***	1.02	1.05	1.07	1.14
2Q.2	0.39 *	1.03	1.05	1.11	1.01	1.03	1.06	1.04	1.10
2Q.3	0.39 *	1.02	1.07	1.12	1.06	1.03	1.04	1.02*	1.04
1Q.1	0.39	1.03	1.06	1.09	1.06	1.07	1.06	1.04	1.04
1Q.2	0.39	1.03	1.06	1.11	1.15	1.16	1.08	1.02	1.05
1Q.3	0.39	1.03	1.06	1.12	1.21	1.22	1.01	1.02	1.06
N.1	0.39	1.03	1.05	1.08	1.04	1.03	1.00	1.05	0.99
N.2	0.39	1.03	1.06	1.11	1.22	1.16	1.00*	1.02	0.98 **
N.3	0.39	1.08	1.07	1.12	1.11*	1.16	0.99 *	1.01*	1.02
B.1	0.39	1.03	1.06	1.10	1.07	1.05	1.03	1.04	1.04
B.2	0.39	1.03	1.01	1.13	1.11	1.22	1.01	1.01	1.01

For further details see the footnote of Table 2

Table 5 Absolute forecast bias over the period 1992Q1–2018Q4

Model	Benchmark			Alternative					
	PM	AR	DFM	MIDAS	LASSO	EN	RS	RP	RF
2Q.1	0.17	1.07	1.03	1.63	0.97	0.96	1.08	1.22	0.85
2Q.2	0.17	1.07	1.04	1.36	1.00	1.05	1.11	1.33	0.92
2Q.3	0.17	1.05	1.05	1.41	1.02	0.97	1.11	1.33	0.88
1Q.1	0.17	1.03	1.06	1.59	0.89	0.92	1.05	1.20	0.84
1Q.2	0.17	1.03	1.10	1.34	0.90	0.95	1.11	1.31	0.92
1Q.3	0.16	0.96	1.14	1.37	1.05	1.03	1.11	1.30	0.82
N.1	0.16	0.95	1.15	1.51	0.93	0.93	1.02	1.15	0.79
N.2	0.16	0.95	1.17	1.31	0.85	0.83	1.08	1.27	0.86
N.3	0.16	0.72	1.13	1.25	1.04	0.93	1.12	1.24	0.81
B.1	0.16	0.98	1.05	1.53	0.83	0.86	1.00	1.15	0.77
B.2	0.16	0.98	0.98	1.29	0.91	0.90	1.06	1.27	0.86

Note: See footnote of Table 2

Table 6 Forecast standard deviation over the period 1992Q1–2018Q4

Model	Benchmark			Alternative					
	PM	AR	DFM	MIDAS	LASSO	EN	RS	RP	RF
2Q.1	0.65	1.03	1.02	1.01	1.00	1.01	0.99	1.01	0.98
2Q.2	0.65	1.03	1.02	1.02	1.00	0.99	1.01	1.00	0.95
2Q.3	0.65	1.03	1.01	1.04	0.99	0.99	0.99	0.99	0.93
1Q.1	0.65	1.03	1.00	0.99	0.99	0.99	1.00	1.00	0.95
1Q.2	0.65	1.03	0.99	1.00	0.99	0.99	0.99	0.98	0.95
1Q.3	0.65	1.03	0.98	0.99	1.01	1.00	0.96	0.97	0.93
N.1	0.65	1.01	0.97	0.96	0.94	0.95	0.95	0.98	0.92
N.2	0.65	1.01	0.94	0.98	0.94	0.92	0.92	0.95	0.90
N.3	0.65	0.97	0.86	0.98	0.86	0.86	0.89	0.91	0.88
B.1	0.65	1.00	0.81	0.92	0.84	0.86	0.90	0.95	0.89
B.2	0.65	1.00	0.77	0.94	0.82	0.86	0.89	0.94	0.88

Note: See footnote of Table 2

weighted forecast combinations and the combinations with weights inversely proportional to past MSFE are extremely similar. The combination scheme with weights based on an expanding window of MSFEs are marginally better than the other two. Compared to the forecasts in Table 2, the combinations provide forecasts that are very close to the model with the lowest MSFE for all horizons. In fact, they are very close to the most precise forecast over all horizons and the second most precise forecast for seven horizons, third most precise forecasts for three horizons and at one horizon close fourth. Using OLS weights, in contrast, results in less precise forecasts

Table 7 Forecast accuracy of forecast combinations

	PM	EA	IP	IP ₁₀	OLS	OLS ₄₀
2Q.1	0.68	1.00*	1.00*	1.00*	1.07	1.12
2Q.2	0.68	0.99*	0.99*	0.99*	1.05	1.14
2Q.3	0.67	0.98*	0.97*	0.97*	1.00	1.08
1Q.1	0.67	0.98	0.98	0.98	1.13	1.23
1Q.2	0.67	0.97	0.97	0.97	1.13	1.19
1Q.3	0.67	0.96*	0.96*	0.96*	1.04	1.09
N.1	0.67	0.94***	0.94***	0.94**	1.05	1.16
N.2	0.67	0.92**	0.92**	0.92**	1.05	1.07
N.3	0.67	0.88	0.87	0.88	0.93	0.94
B.1	0.67	0.87	0.86	0.87	0.89	0.91
B.2	0.67	0.85	0.84	0.84	0.81	0.81

Notes: EA denotes the equally weighted forecast combination, IP forecast combinations with weights inversely proportional to the MSFE, IP₁₀ forecast combinations with weights inversely proportional to the MSFE, where the weights are calculated using a rolling window of 10 quarters, OLS denotes forecast combinations with weights based on the regression of Granger and Ramanathan (1984), and OLS₄₀ denotes forecast combinations with weights based on the regression of Granger and Ramanathan (1984) in a rolling window of 40 quarters. Grey cells indicate the forecast combination with the lowest RMSFE

and nowcasts, which reiterates that forecast biases are not a first order concern in this application.

Forecast combinations using equal and inversely proportional weights, therefore, offers a practical way to ensure that forecasts, nowcasts, and backcasts are very close to the best individual model. This is particularly useful as our results indicate that no individual method provides the best predictions in all circumstances.

4.3 Forecast contributions

Figure 3 to 5 show the contributions of the different time series in each of the five categories to the forecasts of the dynamic factor model, LASSO, random subset regression, and the random forest. We concentrate on these forecasts as they represent the different classes of models and are the respective models with the best forecast accuracy. The top left graph in each figure shows the contributions to the backcast, the top right figure the contribution to the nowcast, the bottom left the contribution to the one quarter ahead forecast, and the bottom right to the two quarter ahead forecast. The contributions are averaged over the months in each quarter to keep the number of plots manageable.

The graphs for the dynamic factor model in Fig. 3 show that the time series in the *Surveys* category had the largest contribution to the forecasts with the relatively largest contribution in the one quarter ahead forecast as relatively more timely survey data arrive. For the nowcasts, the surveys still have the largest, if somewhat

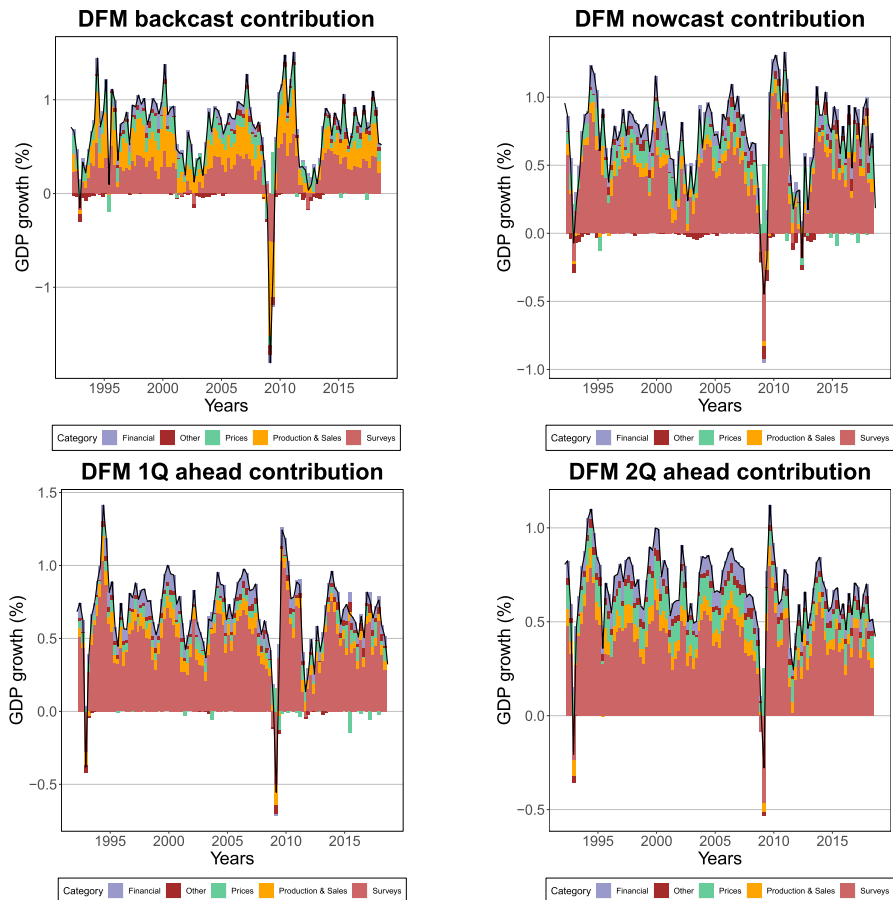


Fig. 3 Contributions of the time series to dynamic factor model forecasts. Note: Category bars indicate the relative part of GDP estimate explained by the five time series categories. The black solid line represents the forecast of GDP by the respective method

smaller, importance and for backcasts the arrival of hard data for the target quarter means that they now contribute more to the backcast. The importance of surveys for nowcasts is in line with the findings of the literature, for example, Giannone et al. (2008) and Bańbura and Rünstler (2011). Gayer and Marc (2018) suggest that the relationship between hard and soft predictors might have changed before and after the Financial Crisis but our results show no such change for the Netherlands.

Figure 4 reports the contributions of the predictor categories when forecasting with the LASSO. Three observations stand out. First, the contributions for the predictions in the first half of the Great Moderation most predictions were based only on the intercept, which is likely due to the low variability of GDP growth during this period. This also explains the results in Table 3, where the LASSO produced results with very similar precision to that of the prevailing mean forecast over the great moderation period.

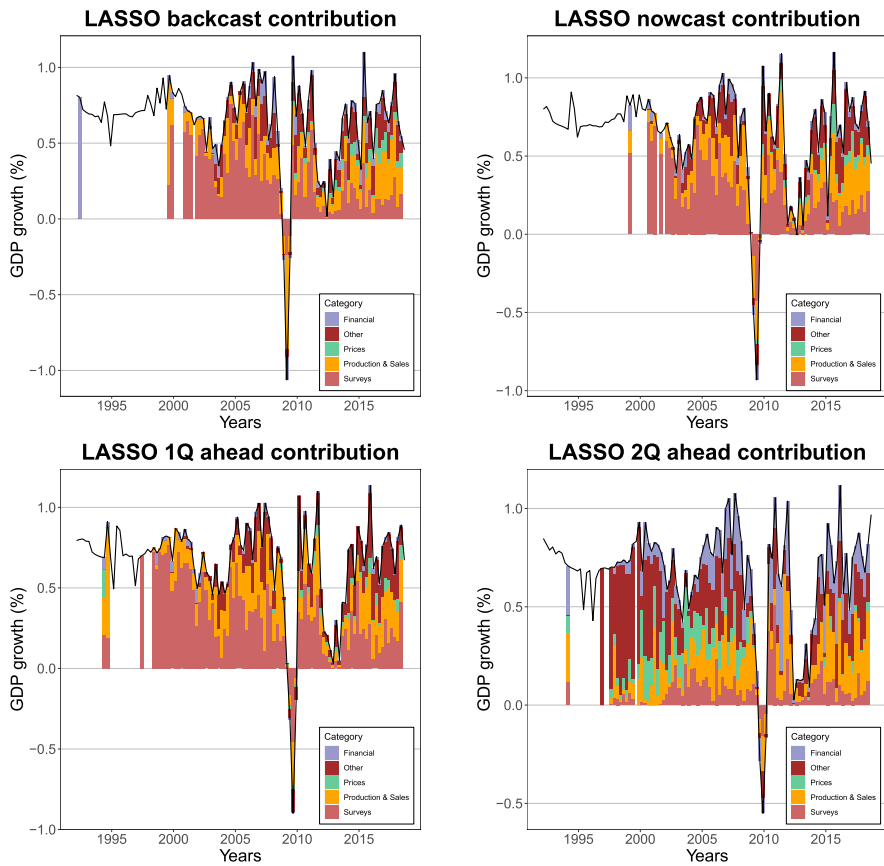


Fig. 4 Contributions of the time series to LASSO forecasts. Note: See footnote of Fig. 3

Second, in the subsequent periods, the LASSO makes more varied use of the information that also fluctuated substantially over time. The two-quarter ahead forecasts makes little use of surveys and relies mainly on hard data. Surveys are more important in one-quarter ahead forecasts, nowcasts and backcasts. However, their importance seemed to decline already before the financial crisis.

Figure 5 displays a more balanced and steady use of the predictors in the different categories. The importance of survey data, which had a sizable but nondominant role before the crisis, remained strong. Predictors in the *Production & Sales* category play an evenly large role and the remaining predictors also contribute throughout. Again, the difference between the forecast horizons is minimal.

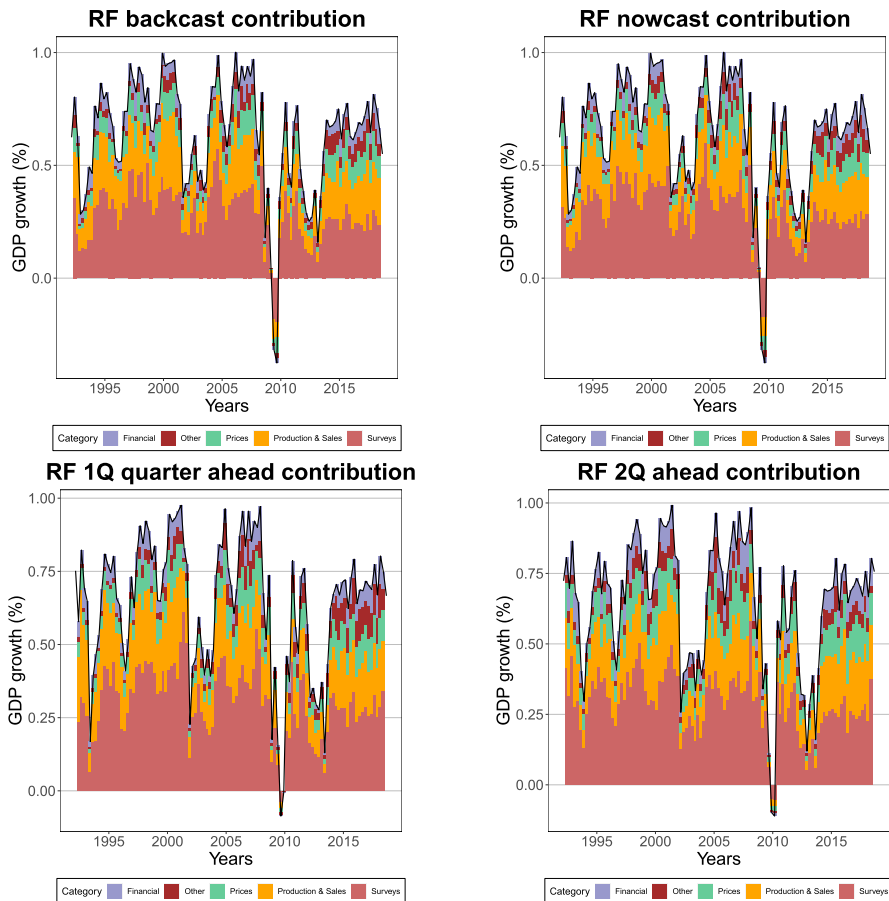


Fig. 5 Contributions of the time series to the random forest forecasts. *Notes:* RF: random forest. See footnote of Fig. 3

5 Conclusion

In this paper, we investigate whether a range of statistical methods often attributed to the machine learning literature can deliver accurate short term forecasts, nowcasts, and backcasts of Dutch GDP.

Our findings suggest that, over the entire forecast period, the random forest delivers the most accurate forecast and nowcasting, whilst the dynamic factor model has the highest forecasting accuracy for backcasting, especially during recessions. Regularization methods perform very well, during the financial crisis in particular. Since, the financial crisis, the random forest has provided the most precise forecast and nowcasts by, for some horizons, substantial margins.

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